EXP.No.: 05
DATE: 24/02/26
PG.No.: 20

DESIGN PCB LAYOUT OF SINGLE-STAGE AMPLIFIER

AIM:

To create a schematic of a single-stage amplifier, design its PCB layout using industry-standard tools (eg:- KiCAD), and prepare the design for fabrication.

SOFTWARE REQUIRED:

- KiCAD or Altium Designer.

PRECAUTIONS:

1. Ensure correct polarity of capacitors.
2. Use heat sinks if required for high-power applications.
3. Check PCB connections before soldering.

PROCEDURE:

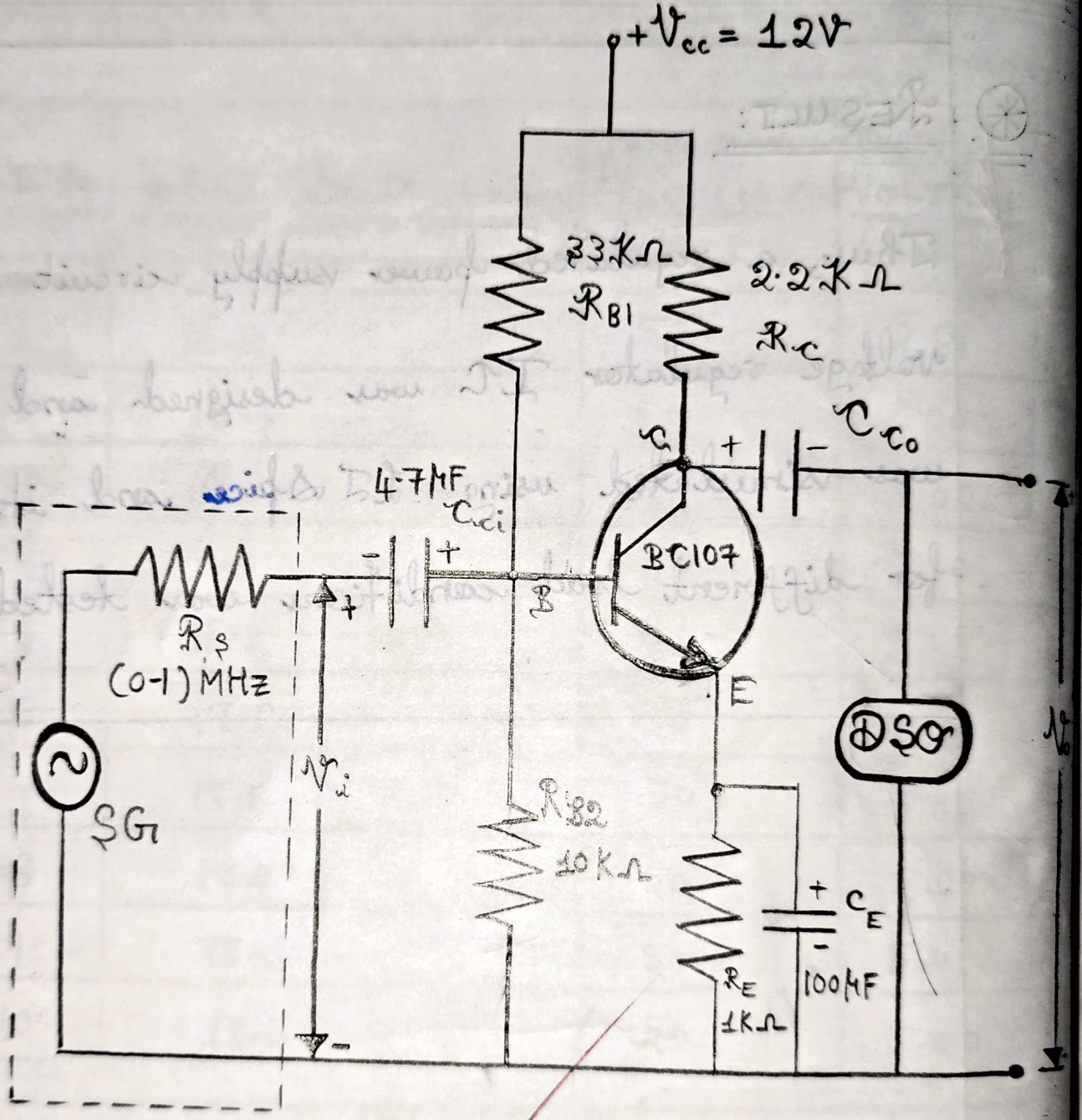
- **Step 1: Design and Simulation:-**

1. Select a transistor (BC107) and determine biasing values.
2. Draw the schematic using simulation software (Multisim, Proteus, or LT Spice).
3. Run the simulation to check the gain and performance.

- **Step 2: Bread board Implementation:-**

1. Assemble the circuit on a bread board.
2. Apply a small AC signal (eg:- 1KHz sine wave).

CIRCUIT DIAGRAM:



3. Measure the output wave-form using an oscilloscope.
4. Verify the gain and functionality.

Step 3: PCB Design:-

1. Open KiCAD/Eagle/Altium and create a new PCB project.
2. Import the schematic and assign footprints to components.
3. Arrange components to minimize noise and enhance performance.
4. Route the PCB tracks, ensuring proper grounding.
5. Generate Gerber files for PCB fabrication.

Step 4: PCB Fabrication:

1. Print the PCB layout on a copper-clad board using a toner transfer method.
2. Etch the board using Ferric Chloride ($FeCl_3$) solution.
3. Drill holes and solder components onto the PCB.

No.: 05
: 24/09/26
.: 22

⊛. OBSERVATIONS:

- Compare the simulated vs practical gain.
- Measure input and output signals using an oscilloscope.
- Observe distortion, noise and stability.

⊛. RESULT:

Thus, the schematic of a single-stage amplifier was designed and its PCB layout was designed using industry-standard tools (eg:- KiCAD) and the design for fabrication was prepared.